

## Appendix D - DS Project Task Planner (Group A)

| Task Board |                                                                                              |                                |                                  |             |   |         |          |                                                                                                                                                                                                                                                                                                                                        |              |              |                   |  |
|------------|----------------------------------------------------------------------------------------------|--------------------------------|----------------------------------|-------------|---|---------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|-------------------|--|
|            | Tasks                                                                                        | Phase                          | Assignee                         | Status      | # | LOE (D) | Priority | Description                                                                                                                                                                                                                                                                                                                            | Start Date   | End Date     | Assessment        |  |
| 1          | Team Contract, roles, tooling setup and finalise Research Topic                              | Project Planning               | All Members                      | Complete    | 1 |         | High     | Agree the working contract and decision rules, confirm roles, and stand up Teams, Planner, OneDrive and the GitHub repository.                                                                                                                                                                                                         | Sep 1, 2026  | Sep 3, 2026  | A1A - Plan        |  |
| 2          | Weekly Lecturer Consultation                                                                 | Project Planning               | All Members                      | In Progress | 7 |         | Medium   | Book the slot, circulate an agenda to the repository 24 hours before, publish minutes within 24 hours after.                                                                                                                                                                                                                           | Sep 3, 2026  | Oct 16, 2026 | All               |  |
| 3          | Project Plan S1: Introduction and Motivation                                                 | Project Planning               | Daniel Cao                       | Complete    | 2 |         | Medium   | Write section 1: the business problem, why medium-term demand forecasting matters, who the client is, and what the project sets out to achieve.                                                                                                                                                                                        | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 4          | Project Plan S2: Brief Literature Review                                                     | Project Planning               | Arman Hajisafi<br>Kelly Hamilton | Complete    | 4 |         | High     | Write section 2: survey existing work on medium- and long-term load forecasting, identify the methodological gap, and state how this project addresses it.                                                                                                                                                                             | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 5          | Project Plan S3: Methods, Software and Data Description                                      | Project Planning               | Faiza Mumtaz<br>Swapnil Shinde   | Complete    | 4 |         | High     | Write section 3: the intended models & why they suit the question, software stack, and a description of the three datasets including format, size and known quality issues.                                                                                                                                                            | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 6          | Project Plan S4: Activities and Schedule                                                     | Project Planning               | Xavier Sun                       | Complete    | 2 |         | Medium   | Write section 4: the activity list, the phase structure, role justification, the timeline and the Gantt chart.                                                                                                                                                                                                                         | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 7          | Group Review Draft - Project Plan                                                            | Project Planning               | All Members                      | Complete    | 1 |         | Medium   | Read the four sections together, resolve overlaps and contradictions, check the three-page limit, and submit.                                                                                                                                                                                                                          | Sep 6, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 8          | Complete Individual Implementation Checklist                                                 | Project Planning               | All Members                      | Complete    | 1 |         | High     | Assessment 1B: Each member's checklist by 5 pm Tuesday of Week 2.                                                                                                                                                                                                                                                                      | Sep 5, 2026  | Sep 8, 2026  | A1B - Checklist   |  |
| 9          | Complete Individual Weekly Self-Reflection                                                   | Reporting & Communication      | All Members                      | In Progress | 4 |         | Medium   | An individual task: write a 500-word summary reflective portfolio, based on your weekly teamwork reflections for Weeks 1-4.                                                                                                                                                                                                            | Aug 31, 2026 | Sep 29, 2026 | A2 - Reflection   |  |
| 10         | Ingest validate and clean datasets                                                           | Data Engineering & Preparation | Daniel Cao<br>Kelly Hamilton     | Not Started | 2 |         | Medium   | Load the three CSVs and confirm columns, types, units, date coverage and row counts against the course brief, check duplicates, and resample temperature onto the 30-minute demand grid.<br>External Data: @Kelly to source Solar-Radiation dataset and dataset clean-up.                                                              | Sep 7, 2026  | Sep 9, 2026  | A3 - Report       |  |
| 11         | Build feature set and design rolling-origin splits                                           | Data Engineering & Preparation | Arman Hajisafi                   | Not Started | 3 |         | Medium   | Calendar and public + school-holiday flags, Fourier seasonality terms, a long-run trend and cooling/heating degree days on a fixed 18 C base, which activity 13 then validates rather than supplies; set the train and test origins for each horizon, then review every feature for information that would not exist at forecast time. | Sep 9, 2026  | Sep 12, 2026 | A3 - Report       |  |
| 12         | Load profiles and seasonality                                                                | Exploratory Data Analysis      | Faiza Mumtaz                     | Not Started | 2 |         | Medium   | Demand by hour, weekday, month and year; STL decomposition and autocorrelation to size the seasonal structure the models have to capture.                                                                                                                                                                                              | Sep 9, 2026  | Sep 11, 2026 | A3 - Report       |  |
| 13         | Temperature–demand response and trend and structural breaks                                  | Exploratory Data Analysis      | Swapnil Shinde                   | Not Started | 3 |         | Medium   | Fit the U-shaped response curve, estimate the heating and cooling thresholds, and quantify how many megawatts each additional degree costs. Quantify the daytime decline driven by rooftop solar across 2010–2021 (isolate the 2020 COVID-19 period).                                                                                  | Sep 9, 2026  | Sep 12, 2026 | A3 - Report       |  |
| 14         | Baseline Modelling                                                                           | Modelling                      | Xavier Sun                       | Not Started | 3 |         | High     | Seasonal naive (same period last year) and a climatology profile averaged over prior years. These set the bar every other model has to clear, and every later result is reported relative to them.                                                                                                                                     | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 15         | Statistical models - SARIMAX                                                                 | Modelling                      | Swapnil Shinde                   | Not Started | 3 |         | High     | Fit SARIMAX with temperature as an exogenous input, with ETS and Prophet as comparators; check residual diagnostics and report the fitted seasonal structure.                                                                                                                                                                          | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 16         | ML Modelling - Random Forest                                                                 | Modelling                      | Daniel Cao                       | Not Started | 3 |         | High     | Train a random forest on the horizon-safe feature set, tuned only on validation origins so the test origins stay untouched; record feature importances for the discussion.                                                                                                                                                             | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 17         | ML Modelling - XGBoost                                                                       | Modelling                      | Faiza Mumtaz                     | Not Started | 3 |         | High     | Train a gradient-boosted tree ensemble with XGBoost on the same features and splits; tune depth, learning rate and regularisation on validation origins only.                                                                                                                                                                          | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 18         | ML Modelling - LightGBM                                                                      | Modelling                      | Arman Hajisafi                   | Not Started | 3 |         | High     | Train a LightGBM ensemble on the same features and splits, and compare its training cost and accuracy against XGBoost, which uses the same algorithm family with a different split-finding strategy.                                                                                                                                   | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 19         | ML Modelling - LSTM                                                                          | Modelling                      | Kelly Hamilton                   | Not Started | 3 |         | High     | Train an LSTM sequence model on the same features and splits, and report whether the extra capacity earns its cost at these horizons.                                                                                                                                                                                                  | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 20         | Model evaluation & validation                                                                | Evaluation & Validation        | All Members                      | Not Started | 2 |         | Medium   | Each member puts their own model through the shared harness: fit at every rolling origin, predict the horizon, collect MAE, RMSE and MAPE. Then break the error down by season, temperature band and demand level, test summer peaks and extreme-heat days.                                                                            | Sep 16, 2026 | Sep 18, 2026 | A3 - Report       |  |
| 21         | Model selection and freeze                                                                   | Evaluation & Validation        | All Members                      | Not Started | 1 |         | Medium   | Compare the six models on the common metrics, choose the best per horizon, record the decision and the evidence behind it, and stop experimenting.                                                                                                                                                                                     | Sep 19, 2026 | Sep 20, 2026 | A3 - Report       |  |
| 22         | Selected model improvement                                                                   | Evaluation & Validation        | Model Winner                     | Not Started | 2 |         | Medium   | The member whose model wins takes one focused round of tuning or feature refinement, with the rest of the team reviewing. The owner is decided at Task 21 (Model Selection).                                                                                                                                                           | Sep 21, 2026 | Sep 23, 2026 | A3 - Report       |  |
| 23         | Report skeleton and section owners, and draft the report chapters                            | Reporting & Communication      | All Members                      | Not Started | 1 |         | Medium   | Agree headings, page budget and a named owner for each of the seven report chapters, then start drafting against the frozen results.                                                                                                                                                                                                   | Sep 21, 2026 | Sep 22, 2026 | A3 - Report       |  |
| 24         | Group Report: Introduction + Conclusion and Further Issues + Organising Reference & Appendix | Reporting & Communication      | Arman Hajisafi                   | Not Started | 5 |         | Medium   | <ul style="list-style-type: none"><li>Chapter 1: Introduction</li><li>Chapter 7: Conclusion and Further Issues</li><li>Reference + Appendix: Sorting &amp; Organising</li></ul>                                                                                                                                                        | Sep 23, 2026 | Oct 7, 2026  | A3 - Report       |  |
| 25         | Group Report: Literature Review + Discussion                                                 | Reporting & Communication      | Kelly Hamilton                   | Not Started | 5 |         | High     | <ul style="list-style-type: none"><li>Chapter 2: Literature Review</li><li>Chapter 6: Discussion</li></ul>                                                                                                                                                                                                                             | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 26         | Group Report: Material and Methods: 3.1, 3.2 and 3.3                                         | Reporting & Communication      | Swapnil Shinde                   | Not Started | 5 |         | Medium   | <ul style="list-style-type: none"><li>Chapter 3:<ul style="list-style-type: none"><li>3.1 Software</li><li>3.2 Description of the Data</li><li>3.3 Pre-processing Steps</li></ul></li></ul>                                                                                                                                            | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 27         | Group Report: Material and Methods: 3.4, 3.5 and 3.6                                         | Reporting & Communication      | Daniel Cao                       | Not Started | 5 |         | High     | <ul style="list-style-type: none"><li>Chapter 3:<ul style="list-style-type: none"><li>3.4 Data Cleaning</li><li>3.5 Assumptions</li><li>3.6 Modelling Methods</li></ul></li></ul>                                                                                                                                                      | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 28         | Group Report: Exploratory Data Analysis                                                      | Reporting & Communication      | Faiza Mumtaz                     | Not Started | 5 |         | Medium   | Chapter 4, written from the EDA notebooks, with every figure regenerated from the cleaned data rather than pasted.                                                                                                                                                                                                                     | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 29         | Group Report: Analysis and Results                                                           | Reporting & Communication      | Xavier Sun                       | Not Started | 5 |         | High     | Chapter 5: the model comparison, the stratified error tables, and what they say about which model wins under which conditions.                                                                                                                                                                                                         | Sep 26, 2026 | Oct 2, 2026  | A3 - Report       |  |
| 30         | Internal review and reproducibility check, and final submission                              | Reporting & Communication      | All Members                      | Not Started | 4 |         | Medium   | Clone the repository fresh, re-run the pipeline end to end, confirm every figure and number in the report regenerates, then submit.                                                                                                                                                                                                    | Sep 29, 2026 | Oct 7, 2026  | A3 - Report       |  |
| 31         | Video storyboard and slides                                                                  | Reporting & Communication      | All Members                      | Not Started | 2 |         | Low      | Storyboard fifteen minutes for a non-technical audience and build one shared deck rather than six.                                                                                                                                                                                                                                     | Oct 7, 2026  | Oct 10, 2026 | A4 - Presentation |  |
| 32         | Record, edit and submit the video                                                            | Reporting & Communication      | Xavier Sun                       | Not Started | 2 |         | Medium   | Each member records their own section; edit the pieces together and submit by 10 pm Tuesday of Week 7.                                                                                                                                                                                                                                 | Oct 10, 2026 | Oct 13, 2026 | A4 - Presentation |  |
| 33         | Project coordination                                                                         | Project Planning               | Xavier Sun                       | In Progress | 4 |         | Medium   | Maintain the task board and Gantt, keep the risk log current, write the weekly agendas and minutes, and enforce the branch-and-review convention on the repository.                                                                                                                                                                    | Aug 31, 2026 | Oct 13, 2026 | All               |  |

+



Appendix E - DS Project Task Planner Timeline - Gantt (Group A)

